

String: fghxx1234xx1234xx1234x1234x56xxx56xxx78xx78xx

subsig: /. \*fgh.\*1234.....56...78/

| pat      | range   |
|----------|---------|
| 1 (fgh)  | (6,∞)   |
| 2 (1234) | (8,∞)   |
| 3 (56)   | (16,16) |
| 4 (78)   | (10,10) |

Input Queue:

| step | loc |
|------|-----|
| 0    | 1   |
| 1    |     |
| 2    |     |
| 3    |     |
| 4    |     |



ToAdd Queue:

| step | loc         |
|------|-------------|
| 0    | 1           |
| 1    | 7           |
| 2    | 19,31,43,53 |
| 3    | 59,69       |
| 4    | 79          |

ToDel Queue:

| step | loc      |
|------|----------|
| 3    | 59       |
| 2    | 19,31,43 |
| 1    |          |



Output Queue

| step | loc |
|------|-----|
| 0    | 1   |
| 1    | 7   |
| 2    | 53  |
| 3    | 69  |
| 4    | 79  |

PatLoc

| pat | id | loc |
|-----|----|-----|
| 1   | 0  | 0   |
| 1   | 1  | 7   |
| 1   | 2  | ∞   |
| 2   | 0  | 0   |
| 2   | 1  | 19  |
| 2   | 2  | 31  |
| 2   | 3  | 43  |
| 2   | 4  | 53  |
| 2   | 5  | ∞   |
| 3   | 0  | 0   |
| 3   | 1  | 59  |
| 3   | 2  | 69  |
| 3   | 3  | ∞   |
| 4   | 0  | 0   |
| 4   | 1  | 79  |
| 4   | 2  | 87  |
| 4   | 3  | ∞   |

Forward Transcript:

| DstPat                                                 | id | locs |
|--------------------------------------------------------|----|------|
| <u>Encoded:</u> SrcStep: 0, SrcLoc: 1, DstRange: (6,∞) |    |      |
| 1                                                      | 0  | 6    |
| 1                                                      | 1  | 7    |
| 1                                                      | 2  | ∞    |
| <u>Encoded:</u> SrcStep: 1, SrcLoc: 7, DstRange: (8,∞) |    |      |
| 2                                                      | 0  | 0    |
| 2                                                      | 1  | 19   |
| 2                                                      | 2  | 31   |
| 2                                                      | 3  | 43   |
| 2                                                      | 4  | 53   |
| 2                                                      | 5  | ∞    |
| ...                                                    |    |      |

Forward Result:

| step | loc         |
|------|-------------|
| 1    | 7           |
| 2    | 19,31,43,53 |
| 3    | 59,69       |
| 4    | 79          |

Backward Branscript:

| PrevPat                                                | loc |
|--------------------------------------------------------|-----|
| <u>Encoded:</u> SrcStep: 4, MinLoc: 79, Range: (10,10) |     |
| 3                                                      | 59  |
| <u>Encoded:</u> SrcStep: 3, MinLoc: 69,Range: (16,16)  |     |
| 2                                                      | 19  |
| 2                                                      | 31  |
| 2                                                      | 43  |
| <u>Encoded:</u> SrcStep: 2, Min: 69 Range: (8,∞)       |     |
| 1                                                      |     |